CM3070 Project
Lecture 1

BSc and Diploma in Computer Science and Related Subjects

Synopsis for Lecture

- Study Plan for Project
- Recap - Overview of CM3070 Project
- Understanding Project Research Ethics
- Recap - Reporting Project Work & Progress
- Developing the Project Proposal (Video)
- Recap – Facilitation & Project Supervision
- Final Things for this Session

Study Plan for Project

Study Structure and Delivery for Project module

- Conducted by Project Leader
  - 1 x 3-hour Overview Workshop
  - 4 x 3-hour Lectures to cover 10 Topics
  - 1 x 3-hour Preparing for Exam Workshop

- Supervisors to provide feedback and review for project deliverables :
  - Through consultation session with students via whatsapp, email or even video chats if preferred.
  - Face-to-face group meetings are also acceptable. Email request to Facilities for a meeting room is allowed.

Useful Info

- We will release the assigned list of project supervisors to students.
- Students who wish for a change of Supervisor may submit a request to students@sim.edu.sg.
- An administrative fee of $54.50 may be imposed to process the request.

Learning Outcomes

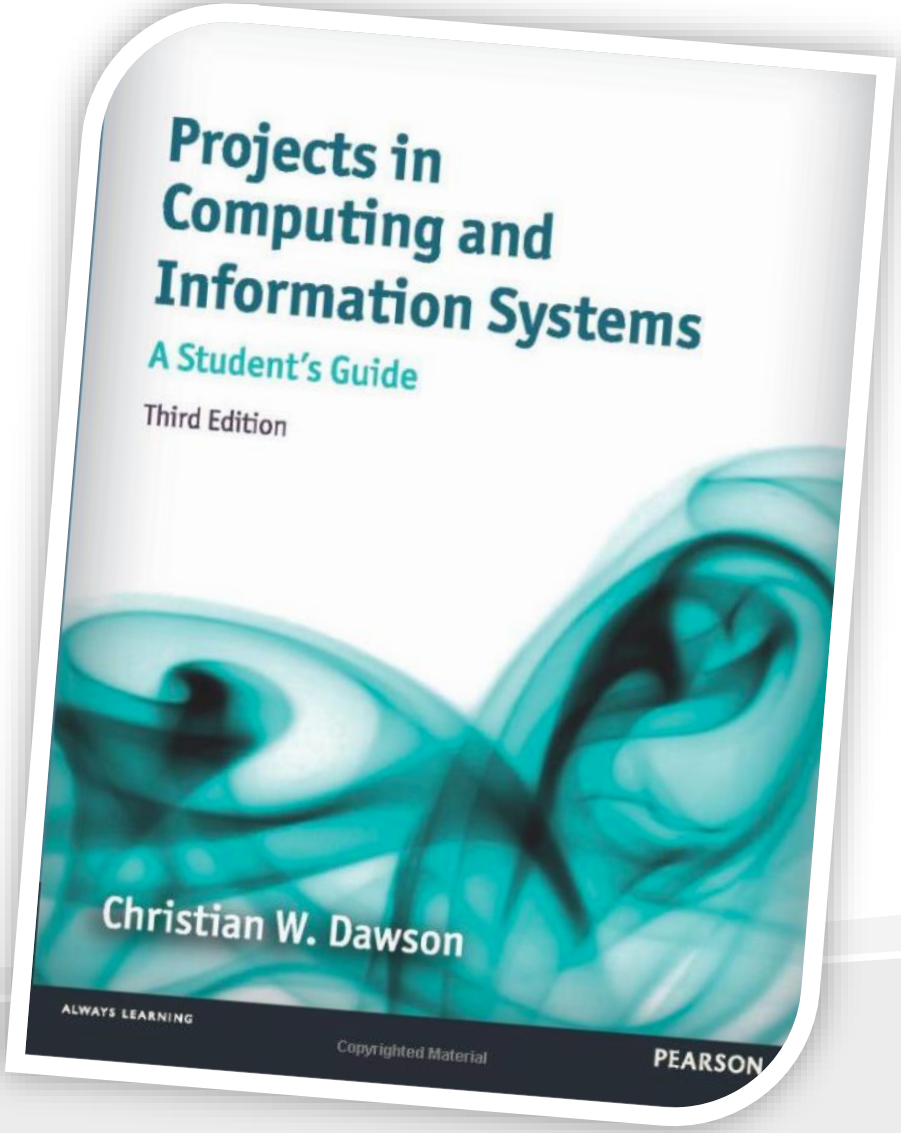
- Students who are enrolled on the BSc Computer Science and on successful completion of the Project, you must be able to:
  - Select and apply appropriate computer science techniques to a particular problem (if **not taking** a specialist pathway)
  - Select and apply appropriate techniques from their specialist area to a particular problem (if **taking** a specialist pathway)
  - Develop a project proposal that can be addressed using computer science techniques
  - Evaluate previous work in areas related to their chosen project, and write a literature review
  - Design and develop a substantial piece of software that matches a project brief
  - Test and evaluate a software project regarding how it matches a project brief including such factors as user needs, software correctness and efficiency
  - Report the results of the project in written and visual form.

Overview of CM3070 **Project**

What is a CM3070 Project ?

- Individual Academic Endeavour.
- Must apply knowledge learnt from earlier modules.
- May use knowledge outside what has been learnt.
- Show elements of analysis, design, development and testing.
- Is more than a copy of case studies from a literature review.
- Must be able to evaluate and show criticism of completed work.

Books/Articles to read …



Research Ethics

- When doing a project, it is very important to consider the ethical issues and impacts that come up.
- Students need to make sure that the project is not harmful. Students are to fill out the quiz (via Coursera), and it will be marked as a pass if the student does not include any ethically problematic issues. See Ethics Checklist in *Coursera Week 3*.
- If students think (though it should very unlikely for this project) there may be some ethical issues with your project, please get in contact with your Project Supervisor to discuss it.

Research with human participants (1/2)

- In many areas of computer science, for example, interaction design, it is important to do research with people.
- It is vital that students make sure that the research does not cause harm or negative consequences such as stress, discomfort or anxiety.
- In some cases, this might not apply to every participant. For example VR studies can cause nausea in some people, that is not a reason to not do them, but you should make sure that people can stop at any time.

Research with human participants (2/2)

- Certain people might be more susceptible to harm, for example if they are pregnant, children, older adults or medical patients. Students should avoid doing research with these populations.
- Another important aspect of working with human participants is that they should be fully informed about what the study involves and give written, signed consent to take part. See sample consent forms in in Coursera Week 3.
- There are also people who are not able to provide free, knowledgeable consent including children, people with learning difficulties or cognitive disabilities, people in legal custody, or other vulnerable groups.

Broader harms

- A research project might cause harm beyond the participants in the research, for example it may cause harm to bystanders (people not directly involved), environmental harm, harm to animals or other physical or societal harms. Some technologies like robotics and artificial intelligence raise particular issues.

Conflict of Interest

- Students should be aware of any potential conflict of interest he / she might have relating to the selected project.
- For example, if the student has any financial involvement in the project, it might make the student less objective about the work.

Protecting Personal Data

- Students should be very careful with the use of personal data in the project, whether student has gathered it, or use third party data.
- Students should ensure that full consent is obtained. In particular, students must consider the following:
  - as a minimum, confidential information should be stored in a password-protected folder on a computer or memory stick; however, password-protected online storage (on OneDrive, Dropbox or similar) is considered more secure
  - any information sheet and consent form should include clear explanations of the data issues identified above, with explicit questions to be answered (as in the templates provided).
  - recordings, images and videos are to be regarded as personal data, and cannot be regarded as anonymised even with voices or faces disguised: these should only be retained with the explicit permission of the participant (discipline-specific and professional codes of conduct will give further guidance).

Reporting Project Work & Progress

Updated as at 21 Sep 2025

PROJECT WORK SCHEDULE (L10B)

The following serves as a guide for students working on the project. Students are advised to work within the schedule, and it is recommended that they put in at least 200 hours of work in order that the project can be completed on time. Additionally, students will have to allocate sufficient time to prepare for the examinations.

SCHEDULE		DATE / TIME
(PRE) Project Workshop 1 – briefing students on • Overview, Expectations, Assessment Criteria, Deliverables • Collection of starting points for ideas for Final Project (two from each level 6 module) • Student will be assigned supervisors by SIM and to be informed at the workshop. Providing an overview & expectations of the CM3070 project unit with tips on submitting a good project proposal (Project Proposal Video).		6 Oct 2025 7pm – 10pm (~1 week before start of semester) HQ BLK B-LT.B.2.05
(L1) Lecture 1 – Topic 1. Project Concept • What is a Project? • Choosing a project idea • Critiquing previous work. Topic 2. Project Proposal • Project proposals • Ethics • Advice on project topics	Learning outcomes: • Select an appropriate topic for a project • Critique related work • Develop a project proposal Learning outcomes: • Identify ethical issues with a project • Select an appropriate scope for a project • Prepare a project proposal	27 Oct 2025 7pm – 10pm HQ BLK B-LT.B.2.05
Producing Project Proposal (Video) – • Students to prepare project proposal (video) • Students to submit project proposal (video) for supervisor's review • Supervisors to complete review and provide feedback to students • Students to clarify issues and refine proposals based on feedback.		UOL Submission deadline: Coursera Week 4
(L2) Lecture 2 – Topic 3. Background Research • Background research • Literature reviews.	Learning outcomes: • Search and discover appropriate literature and previous work • Evaluate previous work and literature • Write a literature review	3 Nov 2025 7pm – 10pm HQ BLK B-LT.B.2.05
(L3) Lecture 3 – Topic 4. Design • Designing your project • Prototyping. Topic 5. Planning and Evaluation • Project Planning • Work plans • Evaluation plans.	Learning outcomes: • Design a project • Critique project designs • Prototype key functionality Learning outcomes: • Plan the development of a substantial project • Plan the evaluation of a project relative to a project brief • Report the results of project design, prototyping and planning	24 Nov 2025 7pm – 10pm HQ BLK B-LT.B.2.05
Producing Preliminary Project Report (PPR) – • Students to write PPR including video for feature prototype, and send to Supervisors for review • Supervisors to provide feedback • Students to clarify issues and refine PPR based on feedback • Students will receive feedback from UOL on project proposal by Coursera Week 7 (tentative)		UOL Submission deadline: Coursera Week 10
• Students to send feedback to Supervisors and work with Supervisors to refine PPR (*students who failed project proposal can re-submit an updated proposal together with PPR).		
(L4) Lecture 4 – Topic 6. Development • Setting up a development environment • Development Advice • Writing up project implementation. Topic 7. Testing and Iteration • Test Criteria • Responding to test results. Topic 8. Academic Writing • Project report structure • Writing a draft report.	Learning outcomes: • Select appropriate computer science techniques for your project • Set up an appropriate development environment • Develop a first iteration of a substantial project Learning outcomes: • Choose appropriate evaluation criteria for your project • Gather feedback from peers • Respond to peer feedback Learning outcomes: • Select appropriate computer science techniques for your project • Set up an appropriate development environment • Develop a first iteration of a substantial project	12 Jan 2026 7pm – 10pm HQ BLK B-LT.B.2.05
Producing Draft Project Report – • Students to prepare draft report and send to Supervisors for review • Supervisors to provide feedback • Students to clarify issues and refine draft report.		UOL Submission deadline: Coursera Week 18
Producing Draft Project Report 2 – • Students to prepare draft report 2 and send to Supervisors for review • Supervisors to provide feedback • Students to clarify issues and refine draft report 2.		UOL Submission deadline: Coursera Week 20
(L5) Lecture 5 – Topic 9. Your project and your career • Computer Science Careers • Portfolios. Topic 10. Completing your project • Advice on completing a project • Developing a demo video • Final project evaluation. – preparing for the Final Exam	Learning outcomes: • Evaluate your project in the context of your future career plans • Present your project as a portfolio piece Learning outcomes: • Prepare your project for final presentation • Evaluate your project as a whole • Create a demo video	9 Feb 2026 7pm – 10pm HQ BLK B-LT.B.2.05
Taking the Project Examination – • Students to study and take Project Examination • Students are expected to have a good understanding of their own project work in order to answer questions in the Project Examination.		UOL Submission deadline: Coursera Week 22
Producing Final Project Report and Code + Project Presentation Video – • Students to prepare presentation video, write progress logs on completed tasks, challenges and issues, etc, and final report • Students to submit to Supervisors for review in Week 20 • Supervisors to provide feedback by Week 22 • Students to clarify issues and refine presentation video, progress logs and final report.		UOL Submission deadline: Coursera Week 24

Plagiarism and Illegality – A Warning

- The work you submit must be your own. Acknowledge any quotations you use from published or unpublished sources.
- Do not copy the work of fellow students or Project Reports completed in previous years. Such unacknowledged quoting or copying (plagiarism) will constitute an examination offence and will be treated as cheating.
- Also, deliberate damage to computer systems or to the work of others, or any misuse of computer systems or acts in contravention of normal legal and ethical requirements may lead to outright failure.

Guidance on the Use of AI in UoL (2025/26) Assessments (7.9 and 7.10) (1/2)

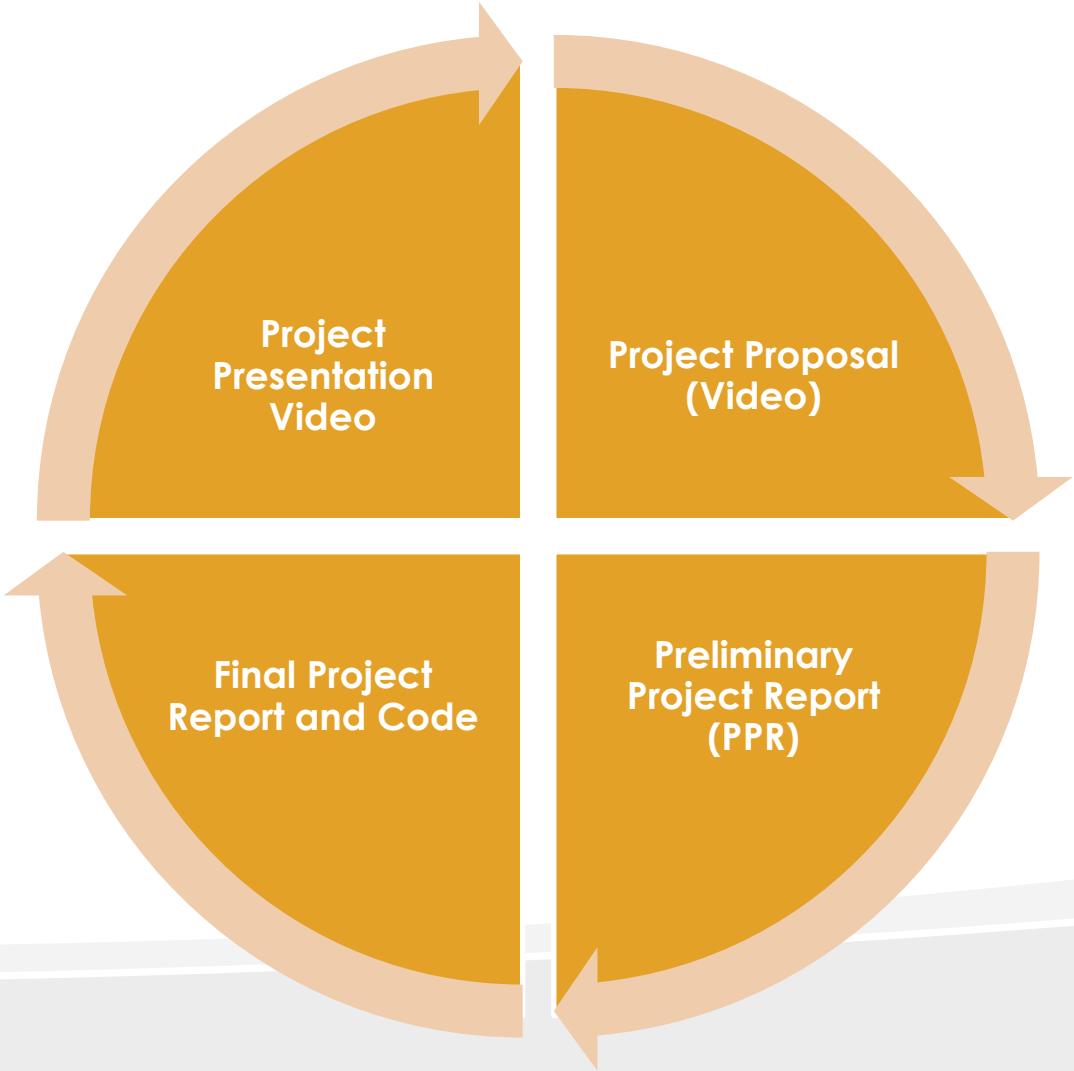
- Academic Misconduct and AI
  - What it is: Submitting work produced by Generative AI is a form of academic misconduct at Goldsmiths, as it is not your own original work.
  - Disclosure is key: Students are encouraged to disclose the use of AI to help establish appropriate citation norms for AI-generated content.
  - All use of AI tools must be acknowledged—either through a statement of use or in-text citations, as directed by the programme team.- Failure to do so is an assessment offence under General Regulation 7.9.
  - **Unless explicitly permitted, students must not submit work generated by Generative AI (e.g., ChatGPT, other AI tools) as their own.**

https://www.gold.ac.uk/media/docs/gam/Academic-Misconduct-Policy-and-Procedures.pdf

Guidance on the Use of AI in UoL (2025/26) Assessments (7.9 and 7.10) (2/2)

- Academic Misconduct and AI (cont'd)
  - Students are responsible for verifying all sources found via AI tools.- Fake or inaccurate references ("hallucinations") must be identified and avoided, as per General Regulation 7.10.
  - **Submitting AI-generated content (in full or part), or content derived from prompts to third-party services, is an offence unless explicitly required or permitted.**
  - Falsifying or fabricating data or information, including visual/audio components, is strictly prohibited.- This includes using AI to create false data unless the assignment specifically requires it (see Regulation 10.1).
  - Consequences: If academic misconduct is identified, Goldsmiths may dismiss the allegation, issue a warning and require submission of the work (appropriately referenced), or take further action

Key Project Deliverables



Project Proposal (Coursera Week 4 Submission)

May 5, 9:00 PM +08

Developing the Project Proposal (1/2)

- The Project Proposal is one of the key deliverables of this Module.
- It doesn't count to student's final grade, but student needs to pass it.
  - If you fail you'll get another opportunity to amend it.
  - Students who failed project proposal can re-submit an updated proposal together with Preliminary Project Report (PPR).
- The submission will be a three to five-minute video presentation explaining the basic concept of the project.

Developing the Project Proposal (2/2)

- This Project Proposal (video) must minimally cover the following areas:
  - The idea for the project
  - Name which template you have used for the project (markers need to cross-refer to the templates)
  - Motivation for the project (why is this an important project?)
  - 2-4 examples of related projects, you should evaluate these examples and use them to explain the motivation for the project (e.g. what is missing in these other projects – requires students to conduct background research)
- A good practice for the video is to use visual supporting elements for your presentation, e.g. PowerPoint or similar slides.

Marking Criteria for Project Proposal

- **Does the report display knowledge of the area of study, previous work and academic literature?** This is those tactful bits of background research.
- **Does it critically evaluate the previous works?** Are you really making that arguments about from the features of that previous work, the benefits, the problems with it, the gaps in that literature to the work that you're doing yourself?
- **Is the project concept justified based on the domain and users?** This is really based on the background research.
- Important Note: See Coursera Week 4 - Critiquing a project proposal for a good check on common issues with student's project proposal.

What makes a good Project Proposal?

Provides the title

Pomodoro Timer

- Short phases of focused attention with rest breaks
- App aimed at online learning students
  - Studying alone: hard to maintain motivation (Diaz 2019)
  - Studying using a web browser: built in distraction (Kumar 2020)

What makes a good Project Proposal?

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Defines the context

What makes a good Project Proposal?

Pomodoro Timer

- Short phases of focused attention with rest breaks
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- Clarifies the aim of project
- Specifies user groups

What makes a good Project Proposal?

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- Demonstrates background research on project idea
- Provides evidence of reading with references

What makes a good Project Proposal?

Pomodoro Timer

- Supporting Motivation
  - Wang and Johnson (2017): visible progress increases motivation
  - Track tasks and check them off
- Web browser distraction
  - Implement as a browser (chrome) extension
  - Block distracting websites during the work periods

- Provides justification of need for project with references

What is missing in this Project Proposal? (1/2)

Pomodoro Timer

- Short phases of focused attention with rest breaks
- App aimed at online learning students
  - Studying alone: hard to maintain motivation (Diaz 2019)
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Pomodoro Timer

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- Web browser distraction
  - Implement as a browser (chrome) extension
  - Block distracting websites during the work periods

- Each bullet point lacks elaboration (assuming that the presenter will elaborate during the video presentation)
- Project Title does not indicate the scope of work
- Name of project idea template is missing
  - Markers cannot cross-refer to the templates
- Definition of Pomodoro Time is without academic reference.

What is missing in this Project Proposal? (2/2)

Pomodoro Timer

- Short phases of focused attention with rest breaks
- App aimed at online learning students
  - Studying alone: hard to maintain motivation (Diaz 2019)
  - Studying using a web browser: built in distraction (Kumar 2020)

Pomodoro Timer

- Supporting Motivation
  - Wang and Johnson (2017): visible progress increases motivation
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- Project Concept is weak without explaining :
  - The features that will be built
  - How the solution can be achieved (processes, techniques and evaluation).
- Examples required were missing :
  - 2-4 examples for comparison to justify the need for the app
- Project development methodology, tools and techniques were not provided
- Deliverables were not elaborated or the lack of mention

Common issues with Students' Project Proposal (1/4)

- **Problem statement is weak.** There was little justification of the need of the project (app). Do note that self-interest as a justification is insufficient and should not be the main motivation behind this conduct of the project.

- A poor example will be:
  - To implement an inventory system because I have interest in this topic.
- A good example will be:
  - To implement an inventory system to automate the manual process of inventory stocktake.

Common issues with Students' Project Proposal (2/4)

- **Lack of background research.** You need to demonstrate that you have conducted research on 2-4 similar apps/model. It should not be simply describing how these existing apps/model work.

- You would need to analyse the
  - Strengths and areas which would be useful for your proposed app/model, and
  - Weaknesses/gaps such that it reinforces the need for your project.

- Elaborate how the gaps could be addressed by your project.

Common issues with Students' Project Proposal (3/4)

- **Weak or lack of academic references.** It is important that credible sources be used when defining certain concepts of your project.

- This is to establish a common understanding as certain terminologies or concepts may not be exactly the same - sometimes due to geographical differences.

- Avoid using non academic references where possible.

Common issues with Students' Project Proposal (4/4)

- **Missing contents.** As part of the proposal, you need to describe how the project can be achieved, i.e. what are the proposed techniques and methods to be used.

- You are also required to describe the features that will be built.

- **Poor structure of proposal.** Contents do not flow smoothly.

Facilitation & Project Supervision

Initial Contact with Your Supervisor (1/2)

<u>Pre-Consultation</u>

- Identify 1 Project Idea which you would like to explore further.
- Develop the proposal (see project proposal requirements) for each idea.
- Do some initial reading on areas where you are likely to work on in the project before contacting your Supervisor.
- The development of the Project Proposal is a careful process– you may decide to deviate in your investigation technique later but this must be well documented

Initial Contact with Your Supervisor (2/2)

Consultation

- The Supervisor will help you narrow the scope of investigation, be able to assess what is "do-able" and give you some directions on where to begin your literature review.
- Discuss with your Supervisor on how you intend to go about collecting information on user requirements or to select methods of investigations.

Post-Consultation

- Decide on "THE" Project Idea and work on the Project Proposal (Video).
- Review and Refine it with the Supervisor.

Initial Contact with Your Supervisor (2/2)

Consultation

- The Supervisor will help you narrow the scope of investigation, be able to assess what is "do-able" and give you some directions on where to begin your literature review.
- Discuss with your Supervisor on how you intend to go about collecting information on user requirements or to select methods of investigations.

Post-Consultation

- Decide on "THE" Project Idea and work on the Project Proposal (Video).
- Review and Refine it with the Supervisor.

Final Things for this Session

Don’t form impressions in your own mind about what is required before **checking with** first your **supervisor** then the **Project Leader**

See you at next session !